ECE 5746 Lab 3 Datasheet

16-bit Registered Multiplier (`mult`), TSMC N16 SVT20

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Design summary

Parameter	Value
Cell library	SVT20 (`BWP20P90`) only, as in Lab 2
# of standard cells (total)	2321
functional (logic, flip-flops, buffers, clock tree)	933
physical (filler, well-tap, boundary)	1388
Macros	0
Core area	$745.563\,\mu m^2$
Chip area	$745.563\,\mu m^2$ (no core-to-die margin)
Die size	$27.54\,\mu m \times 27.07\,\mu m$
Functional cell area	$396.991\,\mu m^2$ (synthesis: $371.796\,\mu m^2$)
Core density (excl. physical cells)	53.2%
clk__per__syn	**0.396 ns** (1.20×0.33 ns; 2525 MHz)
clk__per__apr	**0.7 ns** (1429 MHz)
Critical path delay (post-route)	0.554 ns (`inp_a[7]`→`prod_reg[30]`)
Worst setup slack / hold slack	+0.109 ns / +0.049 ns
Routing	M2–M7, utilization 0.5, total wire length $4241\,\mu m$
Pins	34 inputs on west edge, 32 outputs on east edge

Post-route timing (clock period 0.7 ns)

Setup is checked at the slow corner (ss, 0.72 V, −40 °C, Cworst) and hold at the fast corner (ff, 0.88 V, 125 °C, Cbest). Full reports: `mult.apr_wc3_timing.txt`, `mult.apr_bc3_timing.txt`.

3 worst setup paths				
Path	Start → end	Arrival (ns)	Required (ns)	Slack (ns)
1	`inp_a[7]` → `prod_reg[30]/D`	0.554	0.662	0.109
2	`inp_a[7]` → `prod_reg[31]/D`	0.553	0.662	0.109
3	`inp_a[7]` → `prod_reg[28]/D`	0.553	0.662	0.110
3 worst hold paths				
1	`prod_reg[19]/Q` → `prod[19]`	0.049	0.000	0.049
2	`prod_reg[2]/Q` → `prod[2]`	0.050	0.000	0.050
3	`prod_reg[20]/Q` → `prod[20]`	0.050	0.000	0.050

Finding clk__per__apr

APR used the netlist synthesized at clk__per__syn = 0.396 ns; only the APR clock period in `var.tcl` was changed. From the 0.6 ns run on, Innovus was restricted to SVT20 cells (`setDontUse *LVT*`, `*BWP16P90*`) to match the Lab 2 synthesis; the first run (0.396 ns) had no restriction, and Innovus

swapped most cells to LVT16. A 0.5 ns run without the restriction was also clean, but only because 530 cells were LVT16, so it does not count for SVT20.

APR clock (ns)	Setup WNS (ns)	Cell area (μm^2)	Core density	Result
0.396*	+0.095	676.7	108.6%	core overfilled, cells overlap: not legal
0.6	+0.095	670.7	108.1%	core overfilled, cells overlap: not legal
0.7	**+0.109**	**397.0**	**53.2%**	**clean** (clk_per_apr)
0.8	+0.109	372.2	49.9%	clean

* before the SVT20 restriction (mixed V_t).

At 0.396 and 0.6 ns, Innovus met setup on paper only by upsizing the logic until it no longer fit the core ("Density too high" placement errors, about 200 DRC shorts and open signal nets). Synthesis timed the design at the typical corner with ideal wires and clock; APR times it at the slow corner with routed wires and a real clock tree, so the legal APR clock is longer. 0.7 ns is the shortest period tested with a legal, routed, timing-clean layout.

Verification

- **DRC** (`verify_drc`): 28 violations, all VDD/VSS power stripes extending past the die edge ("Out Of Die", M2–M5) because the floorplan has no core margin.
- **LVS** (`verifyConnectivity`): issues only on power: 92 unconnected well-tap `VPP` pins and VDD/VSS segments; no signal net is open.
- **Post-layout simulation**: `mult.apr.v` with `mult.sdf` (max delays) and the Lab 2 testbench gives all five expected products (0x2, 0xc, 0x0, 0xfffe0001, 0x81b40), each about 0.13–0.15 ns after the clock edge.